

## Chapter 6

# Decision Analysis

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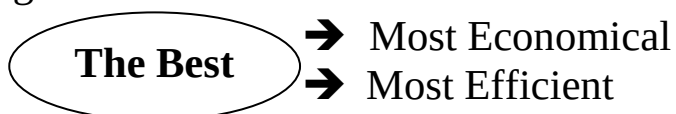
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### 6.1 Introduction

#### 6.1.1 Definition of Decision Making

The process of careful consideration, weighing options, and reaching a conclusion or resolution by an individual or a group to select a specific course of action perceived as the best or the most beneficial among all available alternatives.



#### 6.1.2 Components of a Decision Problem

##### 1. Decision Maker

➔ The individual or entity responsible for making the final decision.

##### 2. Alternatives of Action: $A_j$

➔ These are the actions or strategies that the decision maker must choose to resolve the problem. Since there are typically multiple

options available at any given time, the objective is to select the best possible course of action.

3. Event:  $E_j$

→ These are future events or circumstances that will occur and impact the outcome of the decision. Crucially, these are factors that the decision maker **cannot control**.

4. Payoff:  $C_{ij}$

→ The outcome results from selecting a specific course of action when a particular event occurs. This is typically expressed in the form of Profit, Cost, or Loss.

### 6.1.3 Examples of Decision Problems

**Example 1:** Which course should Daeng enroll in? Between Production Planning and Inventory Control and Simulation.

1. Decision Maker

→ Mr. Daeng

2. Alternatives of Action

→ 1. Enroll in the Planning course      2. Enroll in the Simulation course

3. Event

→ Events are defined by the assigned instructor: either Professor Saowanit or Professor Thitanon.

4. Payoff

→ Grades often depend on the choice of a specific instructor for a course

**Example 2:** A Thai food factory is considering how much to increase production volume next month. Options are: 0%, 10%, or 30%.

1. Decision Maker

→ The executive board of the Thai food factory.

2. Alternatives of Action

→ There are 3 options for increasing production: 0% ( $A_1$ ), 10% ( $A_2$ ), 30% ( $A_3$ )

3. Event

→ The economic conditions for the following month, categorized into 3 states: Recession ( $E_1$ ), Stable ( $E_2$ ), and Boom ( $E_3$ )

#### 4. Payoff

➔ The profit or loss resulting from a specific production increase choice given a particular economic state.

**Example 3:** Engineer Mana invented a new device and secured its patent. A bank provided a loan for him to invest in building his own factory to manufacture the device. He conducted a market survey for the next 5 years and compared the potential returns based on an investment analysis: **If sales are excellent**, he expects a profit of 800,000 Baht/year. If sales are average, he expects 200,000 Baht/year. If sales are poor, he expects a loss of 50,000 Baht/year.

Later, the TOT Company offered to buy his patent along with a profit-sharing agreement. Based on the analysis, if the patent is sold: If sales are excellent, he will earn a profit of 400,000 Baht/year. If sales are average, he will earn 70,000 Baht/year. And, if sales are poor, he will earn 10,000 Baht/year.

#### **Components of this Decision Problem**

##### 1. Decision Maker

➔ Engineer Mana

##### 2. Alternatives of Action

➔ There are 2 choices: Build his own factory ( $A_1$ ) or sell the patent to the TOT Company ( $A_2$ ).

##### 3. Event

➔ Sales conditions over the next 5 years, categorized into 3 states: Excellent sales ( $E_1$ ), Average sales ( $E_2$ ), and Poor sales ( $E_3$ )

##### 4. Payoff

➔ The net profit generated by each alternative under each specific sales condition.

#### **6.1.4 The Decision Making Process**

##### 1. Defining and Analyzing the Problem:

Analyzing and selecting problems identified as critical and requiring urgent resolution.

Problem: A condition where the desired future state deviates from the current or expected reality.

##### 2. Searching, Gathering, and Filtering Information:

- Root Cause of the Problem: Factors that trigger the problem, which are often difficult to see at first glance. They must be studied to ensure the solution targets the correct point.

- Problem Components: Determining which factors have the most severe impact, the duration of that impact, and the areas affected by the problem.

### 3. Determining Alternative Courses of Action:

This process requires the application of academic principles, expertise, experience, creativity, and other relevant factors. In addition, it is necessary to take into account the time and cost associated with identifying and defining these alternatives.

### 4. Evaluating and Comparing Alternatives:

- Pros and cons of each alternative.
- Feasibility of implementation.
- Ease or difficulty of implementation.
- Resource consumption for each option.
- Expected outcomes of implementing that alternative.
- Potential problems and obstacles for each choice.

### 5. Selecting the Best Alternative:



#### Most Economical:

- Choosing the lowest cost
- Minimum production or transportation

costs.



Most Efficient: Any action that achieves specific objectives under a limited budget while generating the maximum result, such as maximum profit, revenue, or output.

**6.1.5 Decision-Making Environments** There are 4 types of decision-making environments, as follows:

#### **1. Decision Making Under Certainty**

This is a situation where the decision maker can accurately and definitively forecast future events or the outcome of each alternative. This is possible because all relevant facts and information are fully known.

**Example:** Nukul is considering investing 500,000 Baht into 3 interesting alternatives as follows:

1. Fixed Deposit Account with a bank: Offering an interest rate of 1.50% with a 3% tax deduction.
2. Fixed Deposit Account with a cooperative: Offering an interest rate of 2.00% with a 3% tax deduction.
3. Government Bonds: With a 1-year redemption period, offering an interest rate of 0.50% and tax-exempt.

## 2. Decision-Making Under Risk

This is a situation where the decision maker must choose among several alternatives while facing multiple possible events. It is impossible to forecast exactly which event will occur; however, the likelihood of each event occurring can be determined in the form of probability.

Determining Probability:

1. Objective probability : Calculated based on historical data of events that occur repeatedly over time.  $\left(\frac{n_i}{n}\right)$

**Example 4:** A 7-Eleven manager wants to study customer demand for detergent to decide how much to order per week to match demand. The sales data and statistics for each week have been collected as follows:

| Sales Volume | Number of Weeks Sold | Probability            |
|--------------|----------------------|------------------------|
| 50           | 8                    | $\frac{8}{40} = 0.200$ |
| 60           | 6                    | $\frac{6}{40} = 0.150$ |
| 70           | 8                    | $\frac{8}{40} = 0.200$ |
| 80           | 7                    | $\frac{7}{40} = 0.175$ |
| 90           | 7                    | $\frac{7}{40} = 0.175$ |
| 100          | 4                    | $\frac{4}{40} = 0.100$ |
| Total        | 40                   | 1                      |

2. Subjective probability : Using judgment or experience to assign the likelihood of each event occurring.

**Example 5:** A company has launched a new product on the market. The market research department relied on experience to help determine the probability values, stating that:

- The probability that the product will be saleable is 80%.
- The probability that the product will not be saleable is 20%.

### **3. Decision Making Under Uncertainty**

This is a situation where the decision maker must choose an alternative without knowing the probability of each event occurring. This happens because past or current data for those events is either unavailable or impossible to obtain.

**Example 6:** A supermarket knows that its profit is affected by the future population growth rate (Low, Normal, High). However, the manager does not know the specific probability of the population growth being at a Low, Normal, or High rate.

### **4. Decision Making Under Competition (Conflict)**

This is a situation where the decision maker must make a choice under uncertainty where the probabilities of various events cannot be determined. Additionally, the decision maker does not know the specific goals or strategies of the opponent (the competitor).

**Example 7:** Football matches, warfare, and business competition.

#### **6.1.6 Payoff Table**

A Payoff Table is a table that displays the resulting outcomes or payoffs ( $C_{ij}$ ) that would be received from selecting each specific alternative under various possible events.

The table shows the return rates for a decision problem with  $n$  options and  $m$  events.

| Action   | Event    |          |          |          |
|----------|----------|----------|----------|----------|
|          | $E_1$    | $E_2$    | ...      | $E_m$    |
| $A_1$    | $C_{11}$ | $C_{12}$ | ...      | $C_{1m}$ |
| $A_2$    | $C_{21}$ | $C_{22}$ | ...      | $C_{2m}$ |
| $\vdots$ |          |          | $\vdots$ |          |
| $A_n$    | $C_{n1}$ | $C_{n2}$ | ...      | $C_{nm}$ |

Payoffs are generally categorized into 3 types as follows.

#### 6.1.6.1 Case: Measuring Payoffs in Terms of Profit

**Example 8:** The Khon Kaen Municipality is deciding whether to build a parking garage in a commercial business district to resolve traffic congestion caused by insufficient parking. A feasibility study identified 3 alternatives:

1. Build a 1-floor building (Capacity: 400 cars).
2. Build a 2-floor building (Capacity: 800 cars).
3. Build a 3-floor building (Capacity: 1,200 cars).

The decision depends on the annual profit, which fluctuates based on the level of demand (High vs. Low). The projected profits are as follows (Baht/year):

- |                                             |                                           |
|---------------------------------------------|-------------------------------------------|
| ● High Demand: $E_2$                        | ● Low Demand: $E_1$                       |
| 1-Floor Building: $A_1$ Profit of 500,000   | 1-Floor Building: $A_1$ Profit of 100,000 |
| 2-Floor Building: $A_2$ Profit of 800,000   | 2-Floor Building: $A_2$ Profit of 60,000  |
| 3-Floor Building: $A_3$ Profit of 1,200,000 | 3-Floor Building: $A_3$ Profit of 20,000  |

**Problem:** Which type of parking garage should the municipality build to maximize annual profit? Construct a Payoff Table.

#### Solution

- Decision Maker  $\Rightarrow$  The Municipal Executive Board.
- Alternatives  $\Rightarrow$  There are 3 options: 1-Floor ( $A_1$ ) 2-Floor ( $A_2$ ) and 3-Floor ( $A_3$ )

- Events  $\Rightarrow$  There are 2 possible events: High Demand ( $E_2$ ) and Low Demand ( $E_1$ )
- Payoff  $\Rightarrow$  The net profit received from each specific alternative.

Payoff Table for the Parking Garage Construction:

| Alternatives               | Events               |                       |
|----------------------------|----------------------|-----------------------|
|                            | Low Demand ( $E_1$ ) | High Demand ( $E_2$ ) |
| 1-Floor Building ( $A_1$ ) | 100,000              | 500,000               |
| 2-Floor Building ( $A_2$ ) | 60,000               | 800,000               |
| 3-Floor Building ( $A_3$ ) | -20,000              | 1,200,000             |

#### 6.1.6.2 Case: Measuring Payoffs in Terms of Costs or Expenses

**Example 9:** Vanich Industry Co., Ltd. manufactures electric fans under its own brand. The company operates two plants: Plant A, located in Rayong, which produces fan components, and Plant B, located in Pathum Thani, which is responsible for assembling the final products. Under normal operations, Plant A delivers components to Plant B in boxes containing 4,000 units each. Based on the experience of the Plant B manager, it is estimated that approximately 5–15% of the components are defective. Given this situation, the manager is considering two alternative courses of action:

1. Inspect every piece sent from Plant A before assembly. If a defective part is found, the inspection cost is 10 Baht per piece.
2. Assemble without inspection. If a fan is found to be non-functional, the defective part is replaced at a cost of 200 Baht per piece.

**Problem:** How should the manager decide if the possible defective rates are 5%, 10%, and 15%?

#### Solution

- Decision Maker  $\Rightarrow$  Manager of Plant B.
- Alternatives  $\Rightarrow$  There are 2 choices: Inspect all pieces ( $A_1$ ), or Do not inspect ( $A_2$ ).

- Events  $\Rightarrow$  There are 3 possible events: Defective rates of 5%, 10%, and 15%.
- Payoff  $\Rightarrow$  Cost

Cost (Expenses) calculated as follows:

If inspecting all pieces: Total cost  $= 10 \times 4,000 = 40,000$  Baht

If not inspected, costs will be incurred for replacing defective parts.

At 5% defective:  $= (5\% \times 4,000) \times 200 = 40,000$  Baht

At 10% defective:  $= (10\% \times 4,000) \times 200 = 80,000$  Baht

At 15% defective:  $= (15\% \times 4,000) \times 200 = 120,000$  Baht

Cost Comparison Table for Different Alternatives (Unit: Baht)

| Alternatives                          | Events       |               |               |
|---------------------------------------|--------------|---------------|---------------|
|                                       | 5% ( $E_1$ ) | 10% ( $E_2$ ) | 15% ( $E_3$ ) |
| Inspect all ( $A_1$ )                 | 40,000       | 40,000        | 40,000        |
| Assemble without inspection ( $A_2$ ) | 40,000       | 80,000        | 120,000       |

### 6.1.6.3 Case: Measuring Payoffs in Terms of Opportunity

#### Loss

There are 2 types of opportunity losses, as follows:

1. Opportunity loss  $\Rightarrow$  This occurs when a product or service is insufficient to meet market demand (Stockout / Shortage). Demand  $>$  Supply

2. Obsolescence loss  $\Rightarrow$  This occurs when the supply of a product or service exceeds market demand (Oversupply / Surplus). Supply  $>$  Demand

Opportunity loss can be calculated in two ways:

1. Based on Profit:  $\Rightarrow$  Profit if the correct decision was made  
 – Profit actually received from that choice.  $\Rightarrow$  Maximum Profit for that Event – Profit of that Alternative.

2. Based on Cost:  $\Rightarrow$  Cost incurred from the chosen alternative  
 – Cost if the correct decision was made.  $\Rightarrow$  Cost of that Alternative – Minimum Cost for that Event.

**Example 10:** Using the data from Example 9, construct an Opportunity Loss Table.

Opportunity Loss Table (*Formula: Alternative Cost - Minimum Cost for that event*)

| Alternatives                          | Events       |               |               |
|---------------------------------------|--------------|---------------|---------------|
|                                       | 5% ( $E_1$ ) | 10% ( $E_2$ ) | 15% ( $E_3$ ) |
| Inspect all ( $A_1$ )                 | 0            | 0             | 0             |
| Assemble without inspection ( $A_2$ ) | 0            | 40,000        | 80,000        |

\* The minimum cost for Event  $E_1$  is 40,000.

The minimum cost for Event  $E_2$  is 40,000.

The minimum cost for Event  $E_3$  is 40,000.

**Example 11:** A 7-Eleven branch is facing a daily problem regarding ordering fresh milk. If the order quantity exceeds customer demand, the milk remains unsold and must be stored or discarded. Conversely, if the order quantity is too low, the store loses the opportunity for additional profit. The store is considering ordering 15, 20, or 25 cases per day. The purchase cost is 150 Baht per case, and the selling price is 250 Baht per case.

Based on experience, daily customer demand is 10, 15, 20, or 25 cases. How many cases should the store order? Construct an Opportunity Loss Table.

### Solution

- Decision Maker  $\Rightarrow$  7-Eleven Store Manager.
- Alternatives  $\Rightarrow$  There are 3 alternatives: ordering 15 ( $A_1$ ), 20 ( $A_2$ ) and 25 ( $A_3$ ) cases per day.
- Events  $\Rightarrow$  There are 4 events: daily customer demand of 10, 15, 20, and 25 cases.
- Payoff  $\Rightarrow$  Net Profit.

In this situation, the loss is calculated based on the profit, by:

Profit on each alternative = Selling price – Cost

- Profit Calculation Examples for Ordering 15 Cases, If Demand = 10:

$$\text{Profit} = \text{Sales} - \text{Cost}$$

$$(250 \times 10) - (150 \times 15) = 250 \text{ Baht}$$

- Profit Calculation Examples for Ordering 15 Cases, If Demand = 15:

$$(250 \times 15) - (150 \times 15) = 1,500 \text{ Baht}$$

- Profit Calculation Examples for Ordering 15 Cases, If Demand = 20:

$$(250 \times 15) - (150 \times 15) = 1,500 \text{ Baht}$$

Profit Table (Unit: Baht)

| Alternatives             | Events             |                    |                    |                    |
|--------------------------|--------------------|--------------------|--------------------|--------------------|
|                          | 10 Cases ( $E_1$ ) | 15 Cases ( $E_2$ ) | 20 Cases ( $E_3$ ) | 25 Cases ( $E_4$ ) |
| Order 15 Cases ( $A_1$ ) | 250                | 1,500              | 1,500              | 1,500              |
| Order 20 Cases ( $A_2$ ) | -500               | 750                | 2,000              | 2,000              |
| Order 25 Cases ( $A_3$ ) | -1,250             | 0                  | 1,250              | 2,500              |

Calculation of Opportunity Loss for ordering 20 cases:

At Demand 10 Cases:  $\Rightarrow (250 \times 10) - (150 \times 20) = -500$

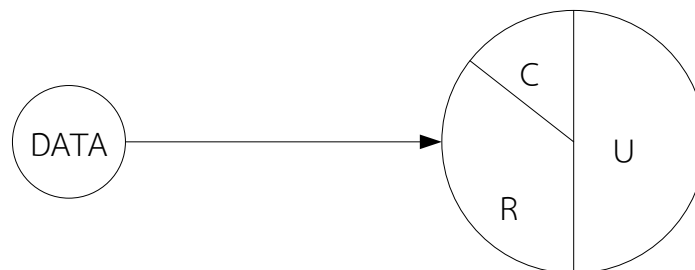
At Demand 25 Cases:  $\Rightarrow (250 \times 10) - (150 \times 20) = 2,000$

Opportunity Loss Table for various alternatives (Unit: Baht) Formula = Maximum Profit – Profit on each alternative

| Alternatives             | Events             |                    |                    |                    |
|--------------------------|--------------------|--------------------|--------------------|--------------------|
|                          | 10 Cases ( $E_1$ ) | 15 Cases ( $E_2$ ) | 20 Cases ( $E_3$ ) | 25 Cases ( $E_4$ ) |
| Order 15 Cases ( $A_1$ ) | 0                  | 0                  | 500                | 1,000              |
| Order 20 Cases ( $A_2$ ) | 750                | 750                | 0                  | 500                |
| Order 25 Cases ( $A_3$ ) | 1,500              | 1,500              | 750                | 0                  |

**6.1.7 Types of Decision Models** Decision models are categorized into 3 types as follows:

1. Decision Making Under Certainty ( $C$ )
2. Decision-Making Under Risk ( $R$ ): The probability (Prob) is known.
3. Decision Making Under Uncertainty (Uncertainty;  $U$ ): The probability (Prob) is unknown



### 6.1.8 Single-Stage and Multi-Stage Decision Making

#### 6.1.8.1 Single-Stage Decision Making

**Example 12:** A 7-Eleven branch is facing a daily problem regarding ordering fresh milk. If the quantity ordered exceeds customer demand, there will be a surplus that the store must either keep or discard. Conversely, if the quantity ordered is less than demand, the store loses the opportunity for additional profit. The store is currently considering daily order quantities of 15, 20, or 25 cases. The purchase cost is 150 Baht per case, and the selling price is 250 Baht per case. Based on past experience, daily customer demand varies between 10, 15, 20, and 25 cases. How many cases should the store order?

#### 6.1.8.2 Multi-Stage Decision Making

**Example 13:** Tech Company, an automotive components manufacturer, has learned that a small developing country is interested in purchasing 2,500 units of carburetors. The company has been invited to submit a proposal that includes a specified delivery schedule. If the country is interested, a contract will be signed; otherwise, no agreement will be made. Consequently, the company faces two initial alternatives: to prepare a proposal or to ignore the invitation.

If the company submits a proposal and is awarded the contract, the subsequent challenge is whether they can meet the government's production specifications. Due to manufacturing constraints, they must decide whether to source some parts from other suppliers or to manufacture new units to ensure timely delivery.

Based on this information, how should the Company President proceed to resolve this problem?

### Solution

- Decision Maker  $\Rightarrow$  The Company President.
- Alternatives  $\Rightarrow$  There are 6 primary alternatives:

Define:  $A_1$  Terminate the project.

$A_2$  Prepare the project.

$A_3$  Do not manufacture carburetors.

$A_4$  Manufacture carburetors.

$A_5$  Purchase 2,500 units of carburetors.

$A_6$  Manufacture 2,500 units of carburetors.

- Events  $\Rightarrow$  There are 4 events, namely:

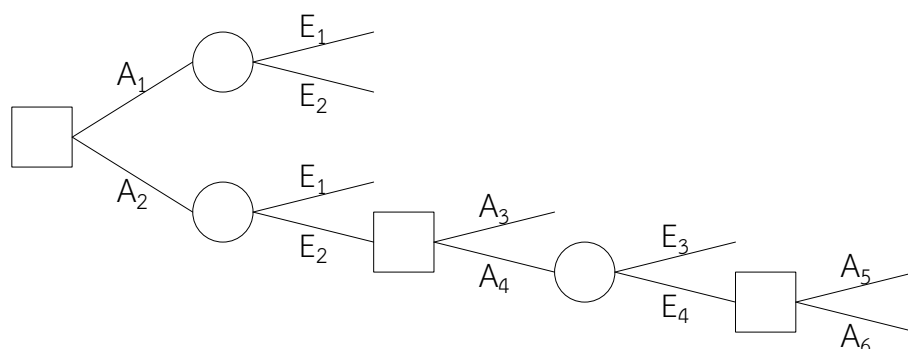
Define:  $E_1$  The government does not award the contract.

$E_2$  The government awards the contract.

$E_3$  No defective carburetors.

$E_4$  Defective carburetors at p %

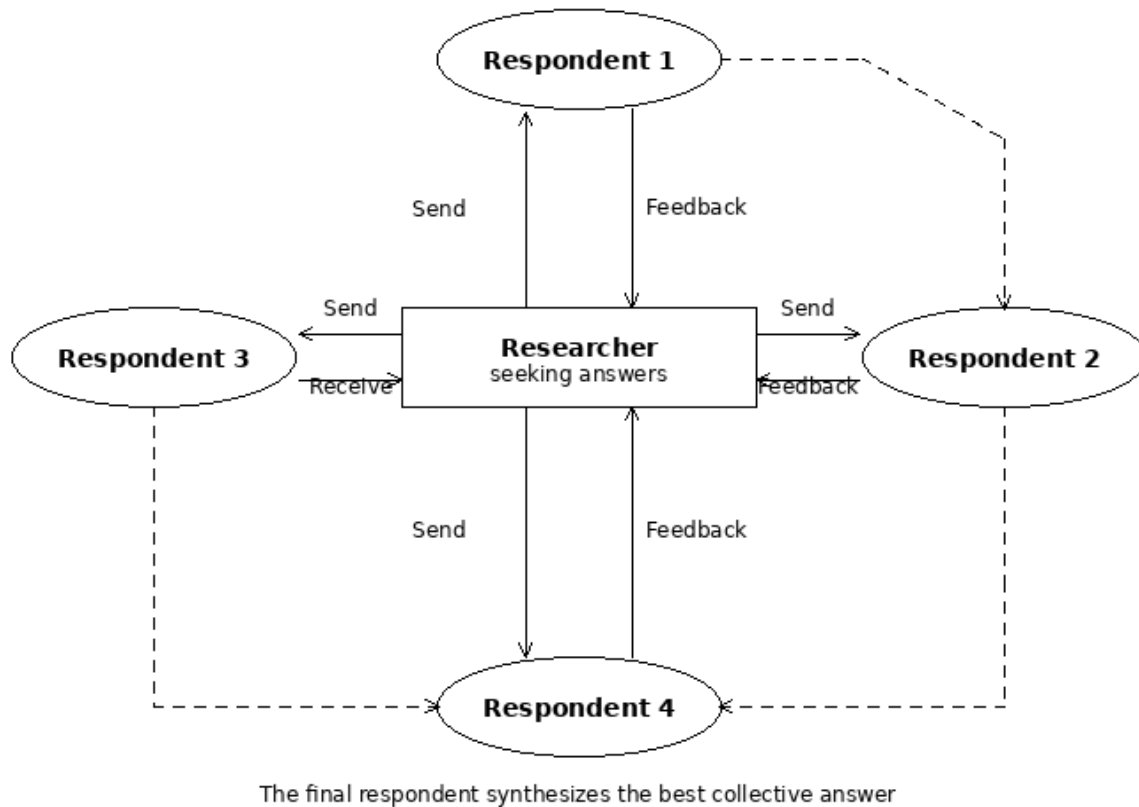
- Payoff  $\Rightarrow$  Profit received



**6.1.9 Decision-Making Strategies:** are categorized into 2 main groups as follows:

1. Qualitative Techniques

- Judgment and Experience
- Qualitative Research (Focus group)
- Delphi Technique



## 2. Quantitative Techniques

- Probability
- Linear Programming
- Network Models
- Simulation
- Queuing Theory

## 6.2 Decision Making Under Uncertainty

### 6.2.1 Decision Making Under Certainty

**Example 14:** Wimonsiri is considering investing 1 million Baht in three attractive alternatives:

1. Fixed Deposit with a bank: Interest rate of 1.50%, subject to 3% tax.
2. Fixed Deposit with a cooperative: Interest rate of 2.00%, subject to 3% tax.
3. Government Bonds: 1-year maturity, interest rate of 0.50%, tax-exempt.

Which alternative should Wimonsiri choose for the best return?

### 6.2.2 Decision Making Under Uncertainty

This refers to situations where the decision maker must choose an alternative without knowing the probability of each event occurring. This is due to a lack of available historical or current data regarding those events.

### 6.2.3 Decision-Making Criteria (Strategies)

- MAXIMIN Criterion
- MAXIMAX Criterion
- MINIMAX Criterion
- HURWICZ Criterion
- LAPLACE Criterion
- EMV Criterion

**Example 15:** An instant noodle company is deciding between 3 marketing strategies for a new product:

Strategy A: Buy 1 Get 1 Free.

Strategy B: A chance to find gold (1 Salung weight) inside the package.

Strategy C: Collect the letters "MAMA" to win 1 million Baht cash.

The company anticipates three levels of consumer acceptance: **High, Moderate, and Low**. The projected profits (in millions of Baht) for each strategy and event are as follows:

| Strategy   | Expected event (in million baht) |       |       |
|------------|----------------------------------|-------|-------|
|            | $E_1$                            | $E_2$ | $E_3$ |
| Strategy A | 14                               | 12    | 1     |
| Strategy B | 3                                | 3     | 4     |
| Strategy C | 8                                | 2     | 9     |

#### 6.2.3.1 MAXIMIN Criterion

1. Identify the minimum payoff for each alternative (applicable to profit tables).

2. Select the alternative with the maximum value among those minimum payoffs identified in Step 1.

**Example 16:** From Example 15, Constructing the Payoff Table as follows (Unit: Million Baht):

| Strategy   | Expected event |       |       | Minimum Payoff |
|------------|----------------|-------|-------|----------------|
|            | $E_1$          | $E_2$ | $E_3$ |                |
| Strategy A | 14             | 12    | 1     | 1              |
| Strategy B | 3              | 3     | 4     | 3              |
| Strategy C | 8              | 2     | 9     | 2              |

⇒ Select Strategy B.

#### 6.2.3.2 MAXIMAX Criterion

1. Identify the maximum payoff for each alternative (applicable to profit tables).

2. Select the alternative with the highest value among the maximum payoffs identified in Step 1.

**Example 17:** From Example 15, Constructing the Payoff Table as follows (Unit: Million Baht):

| Strategy   | Expected event |       |       | Maximum Payoff |
|------------|----------------|-------|-------|----------------|
|            | $E_1$          | $E_2$ | $E_3$ |                |
| Strategy A | 14             | 12    | 1     | 14             |
| Strategy B | 3              | 3     | 4     | 4              |
| Strategy C | 8              | 2     | 9     | 9              |

⇒ Select Strategy A.

#### 6.2.3.3 MINIMAX Criterion

1. Identify the maximum opportunity loss (regret) for each alternative.
2. Select the alternative with the **minimum value** among the maximum opportunity losses identified in step 1.

**Example 18:** From Example 15, Constructing the Opportunity Loss Table (Unit: Million Baht):

| Strategy   | Expected event |       |       | Maximum Opportunity Loss |
|------------|----------------|-------|-------|--------------------------|
|            | $E_1$          | $E_2$ | $E_3$ |                          |
| Strategy A | 0              | 0     | 8     | 8                        |
| Strategy B | 11             | 9     | 5     | 11                       |
| Strategy C | 6              | 10    | 0     | 10                       |

⇒ Select Strategy A.

#### 6.2.3.4 HURWICZ Criterion

Calculate the average payoff for each alternative using the following formula:

$$EA_i = \alpha \times (\text{Maximum payoff}) + (1 - \alpha) \times (\text{Minimum payoff})$$

Where  $\alpha$  (alpha) is the Coefficient of Optimism, ranging from 0 to 1:

$\alpha = 1 \Rightarrow$  Completely optimistic.

$\alpha = 0 \Rightarrow$  Completely pessimistic.

$\alpha = 0.25 \Rightarrow$  25% optimistic.

The  $EA_i$  value will be chosen as Max or Min depending on the type shown in the payoff table. If

it is in the form of profit  $\Rightarrow EA_i : \text{Max}$

it is in the form of cost  $\Rightarrow EA_i : \text{Min}$

**Example 19** From Example 15, apply the Hurwicz criterion when  $\alpha = 0.25$ , the payoff table is as follows (Unit: Million Baht).

| Strategy   | Expected event |       |       | Average payoff              |
|------------|----------------|-------|-------|-----------------------------|
|            | $E_1$          | $E_2$ | $E_3$ |                             |
| Strategy A | 14             | 12    | 1     | $0.25(14) + 0.75(1) = 4.25$ |
| Strategy B | 3              | 3     | 4     | $0.25(4) + 0.75(3) = 3.25$  |
| Strategy C | 8              | 2     | 9     | $0.25(9) + 0.75(2) = 3.75$  |

$\Rightarrow$  Select Strategy A.

If it's a loss table:

$$EA_i = \alpha \times (\text{Minimum payoff}) + (1 - \alpha) \times (\text{Maximum payoff})$$

$$EA_1 = 0.25(0) + 0.75(8) = 6$$

$$EA_2 = 0.25(5) + 0.75(11) = 9.5$$

$$EA_3 = 0.25(0) + 0.75(10) = 7.5$$

#### 6.2.3.5 LAPLACE Criterion

1. Calculate the average payoff for each alternative by assuming all events have an equal probability of occurring.

$$E(A_j) = \sum_{i=1}^m C_{ij} P(S_i)$$

2. Select the alternative with the highest average payoff.

**Example 20:** From Example 15, the payoff table is as follows (Unit: Million Baht).

| Strategy    | Expected event |               |               | Average payoff                          |
|-------------|----------------|---------------|---------------|-----------------------------------------|
|             | $E_1$          | $E_2$         | $E_3$         |                                         |
| Strategy A  | 14             | 12            | 1             | $\frac{14+12+1}{3} = \frac{27}{3} = 9$  |
| Strategy B  | 3              | 3             | 4             | $\frac{3+3+4}{3} = \frac{10}{3} = 3.33$ |
| Strategy C  | 8              | 2             | 9             | $\frac{8+2+9}{3} = \frac{19}{3} = 6.33$ |
| Probability | $\frac{1}{3}$  | $\frac{1}{3}$ | $\frac{1}{3}$ |                                         |

⇒ Select Strategy A.

#### 6.2.3.6 EMV (Expected Monetary Value) Criterion

1. Calculate the average payoff for each alternative by assigning probabilities to each state of nature based on the decision maker's beliefs. The formula is as follows:  $E(x) = \sum_{i=1}^n x_i P(x)$

2. Select the alternative with the highest average payoff.

**Example 21:** From Example 15, the decision maker assigns probabilities to each state of nature, and the payoff table is as follows (Unit: Million Baht).

| Strategy    | Expected event |               |               | Average payoff                                                                                |
|-------------|----------------|---------------|---------------|-----------------------------------------------------------------------------------------------|
|             | $E_1$          | $E_2$         | $E_3$         |                                                                                               |
| Strategy A  | 14             | 12            | 1             | $14\left(\frac{1}{2}\right) + 12\left(\frac{1}{4}\right) + 1\left(\frac{1}{4}\right) = 10.25$ |
| Strategy B  | 3              | 3             | 4             | $3\left(\frac{1}{2}\right) + 3\left(\frac{1}{4}\right) + 4\left(\frac{1}{4}\right) = 3.25$    |
| Strategy C  | 8              | 2             | 9             | $8\left(\frac{1}{2}\right) + 2\left(\frac{1}{4}\right) + 9\left(\frac{1}{4}\right) = 6.75$    |
| Probability | $\frac{1}{2}$  | $\frac{1}{4}$ | $\frac{1}{4}$ |                                                                                               |

⇒ Select Strategy A.

**Summary:** MAXIMIN yields a result that differs from the other methods, which mostly favor Alternative A. Therefore, Alternative A should be chosen.

## 6.3 Decision-Making Under Risk

### 6.3.1 Expected Value

Suppose there are  $n$  alternatives and  $m$  possible states of nature. The expected payoff from alternative  $i$  ;  $EV(A_i)$  can be calculated as:

$$EV(A_i) = \sum_{j=1}^m C_{ij} \Pr(E_j)$$

Where:  $\Pr(E_j)$  = the probability that state of nature  $j$  will occur.

$C_{ij}$  = the payoff from alternative  $i$  when state of nature  $j$  occurs.

**Example 22:** A company is considering two investment projects, Project A and Project B, requiring the same amount of investment. The profits of both projects depend on future economic conditions. The payoff information is summarized in the table below:

| Economic Condition | Probability | Alternatives        |                     |
|--------------------|-------------|---------------------|---------------------|
|                    |             | Project A ( $A_1$ ) | Project B ( $A_2$ ) |
| $E_1$ : Recession  | 0.2         | 400 ( $C_{11}$ )    | 0 ( $C_{12}$ )      |
| $E_2$ : Normal     | 0.6         | 500 ( $C_{21}$ )    | 500 ( $C_{22}$ )    |
| $E_3$ : Boom       | 0.2         | 600 ( $C_{31}$ )    | 1000 ( $C_{32}$ )   |

Calculate the expected payoff for each alternative.

### Solution

The expected payoff for each alternative is calculated using the formula:

$$EV(A_i) = \sum_{j=1}^m C_{ij} \Pr(E_j); \text{ since } C_{ij} \text{ is expressed in terms of profit}$$

$$\Rightarrow EV(A_1) = 0.2(400) + 0.6(500) + 0.2(600) = 500$$

$$EV(A_2) = 0.2(0) + 0.6(500) + 0.2(1000) = 500$$

$\Rightarrow$  Which alternative yields the highest profit?

### 6.3.2 Expected Opportunity Loss (EOL)

The formula used is the same as for expected value, except that the  $C_{ij}$  is expressed in terms of loss (regret) rather than profit.

$$EOL(A_i) = \sum_{j=1}^m C_{ij} \Pr(E_j)$$

\*Choose the alternative with the lowest EOL.

### 6.3.3 Risk

Risk refers to a situation in which the actual outcome may vary or deviate from the expected outcome.

**Example 23** Nipon is considering investing **1 million baht** in one of two alternatives:

1. A fixed deposit with a cooperative offering 2% interest and pay 3% tax.

2. Buying DTAC stock, expected to yield a 10% annual return.

Risk calculation formula:

$$R_i = \sqrt{\sum_{j=1}^m [C_{ij} - EV(A_i)]^2 \Pr(E_j)}$$

Where:  $C_{ij}$  = the payoff obtained from alternative  $i$  when event  $j$  occurs

$EV(A_i)$  = the expected payoff of the alternative  $i$

$\Pr(E_j)$  = the probability that event  $j$  will occur

$R_i$  = the risk of the alternative  $i$

**Example 24:** From Example 22, calculate the risk of each alternative.

**Solution** From the previous results  $EV(A_1) = 500$ ,  $EV(A_2) = 500$ , the expected values are equal, so we must compare the **risk** of each alternative. After calculating the risk values:

$$R_i = \sqrt{\sum_{j=1}^m [C_{ij} - EV(A_i)]^2 \Pr(E_j)}$$

$$R_1 = \sqrt{(400 - 500)^2(0.2) + (500 - 500)^2(0.6) + (600 - 500)^2(0.2)} = 63.25$$

$$R_2 = \sqrt{(0 - 500)^2(0.2) + (500 - 500)^2(0.6) + (1000 - 500)^2(0.2)} = 316.23$$

$\Rightarrow$  Choose Project A because it has a lower risk.

- If risks are equal  $\Rightarrow$  choose the alternative with the higher expected profit (EV) or the lower expected opportunity loss (EOL).

- If the expected values (EV or EOL) are equal  $\Rightarrow$  choose the alternative with the lower risk.

- If both expected return and risk are not equal  $\Rightarrow$  use  $C.V. = \frac{\sigma}{\mu} \times 100 = \frac{R_i}{EV(A_i)}$

- If the C.V. values are equal, choose the alternative with the higher expected return.

### 6.3.4 Coefficient of Variation (CV) of Risk

The coefficient of variation of risk for alternative  $i$ .

$$C.V.(R_i) = \frac{R_i}{EV(A_i)}$$

**Example 25:** From Examples 22 and 24, calculate the coefficient of variation of risk for each alternative.

We obtain:  $C.V.(R_1) = \frac{R_1}{EV(A_1)} = \frac{63.25}{500} = 0.1265$

$$C.V.(R_2) = \frac{R_2}{EV(A_2)} = \frac{316.25}{500} = 0.6325$$

⇒ Choose Project A.

### 6.3.5 Strategies for Decision-Making Under Risk

#### - Decision-making under risk

refers to situations where the decision maker knows the available alternatives but does not know with certainty which outcome will occur. However, they are able to estimate or assign probabilities to the possible events.

- Maximum likelihood method
- Maximum expected profit method
- Minimum expected opportunity loss method
- Minimum risk method

**Example 26:** A company that produces a brand of instant noodles is deciding among three marketing strategies to increase sales:

Strategy A: Buy 1 Get 1 Free.

Strategy B: A gold prize (1 salung in weight) inside the package.

Strategy C: Collect the letters spelling “MAMA” to win 1 million baht in cash.

The company anticipates events that may influence consumer acceptance, namely  $E_1$ ,  $E_2$ ,  $E_3$ , and estimates the profits to be earned under each strategy and each event as follows:

| Strategy    | Expected events (Unit: Million Baht) |       |       |
|-------------|--------------------------------------|-------|-------|
|             | $E_1$                                | $E_2$ | $E_3$ |
| Strategy A  | 14                                   | 12    | 1     |
| Strategy B  | 3                                    | 3     | 4     |
| Strategy C  | 8                                    | 2     | 9     |
| Probability | 0.2                                  | 0.6   | 0.2   |

Determine the most appropriate alternative using the different methods.

#### 6.3.5.1 Maximum Likelihood Method

1. Select the event with the highest probability of occurrence.
2. Choose the alternative that provides the highest payoff under that event.

**Solution**

| Strategy    | Expected Events |       |       |
|-------------|-----------------|-------|-------|
|             | $E_1$           | $E_2$ | $E_3$ |
| Strategy A  | 14              | 12    | 1     |
| Strategy B  | 3               | 3     | 4     |
| Strategy C  | 8               | 2     | 9     |
| Probability | 0.2             | 0.6   | 0.2   |

⇒ Choose Strategy A.

**6.3.5.2 Maximum Expected Profit Method**

Choose the alternative with the highest expected payoff.

**Solution**

| Strategy    | Expected Events |       |       | Expected Payoff $EV(A_i)$           |
|-------------|-----------------|-------|-------|-------------------------------------|
|             | $E_1$           | $E_2$ | $E_3$ |                                     |
| Strategy A  | 14              | 12    | 1     | $0.2(14) + 0.6(12) + 0.2(1) = 10.2$ |
| Strategy B  | 3               | 3     | 4     | $0.2(3) + 0.6(3) + 0.2(4) = 3.2$    |
| Strategy C  | 8               | 2     | 9     | $0.2(8) + 0.6(2) + 0.2(9) = 4.6$    |
| probability | 0.2             | 0.6   | 0.2   |                                     |

⇒ Choose Strategy A.

**6.3.5.3 Minimum Expected Opportunity Loss Method**

Choose the alternative with the lowest expected opportunity loss.

**Solution**

| Strategy    | Expected Events |       |       | $EOL(A_i)$                        |
|-------------|-----------------|-------|-------|-----------------------------------|
|             | $E_1$           | $E_2$ | $E_3$ |                                   |
| Strategy A  | 0               | 0     | 8     | $0.2(0) + 0.6(0) + 0.2(8) = 1.6$  |
| Strategy B  | 11              | 9     | 5     | $0.2(11) + 0.6(9) + 0.2(5) = 8.6$ |
| Strategy C  | 6               | 10    | 0     | $0.2(6) + 0.6(10) + 0.2(0) = 7.2$ |
| Probability | 0.2             | 0.6   | 0.2   |                                   |

⇒ Choose Strategy A.

**6.3.5.4 Minimum Risk Method**

Choose the alternative with the lowest risk or the lowest  $CV.(R_i)$ .

### Solution

| Strategy    | Expected Events |       |       | $E_3$  | $C.V.(R_i)$             |
|-------------|-----------------|-------|-------|--------|-------------------------|
|             | $E_1$           | $E_2$ | $E_3$ |        |                         |
| Strategy A  | 14              | 12    | 1     | 4.6647 | $4.6647 / 10.2 = 0.457$ |
| Strategy B  | 3               | 3     | 4     | 0.4    | $0.4 / 3.2 = 0.125$     |
| Strategy C  | 8               | 2     | 9     | 3.2    | $3.2 / 4.6 = 0.696$     |
| Probability | 0.2             | 0.6   | 0.2   |        |                         |
| $EV(A_i)$   | 10.2            | 3.2   | 4.6   |        |                         |

⇒ Choose Strategy B

By

$$R_1 = \sqrt{(14 - 10.2)^2(0.2) + (12 - 10.2)^2(0.6) + (1 - 10.2)^2(0.2)} = \sqrt{21.76} = 4.6647$$

$$R_2 = \sqrt{(3 - 3.2)^2(0.2) + (3 - 3.2)^2(0.6) + (4 - 3.2)^2(0.2)} = \sqrt{0.16} = 0.4$$

$$R_3 = \sqrt{(8 - 4.6)^2(0.2) + (2 - 4.6)^2(0.6) + (9 - 4.6)^2(0.2)} = \sqrt{10.24} = 3.2$$

### 6.3.6 Expected Value of Perfect Information (EVPI)

⇒ EVPI represents the value of perfect information.

⇒ It helps determine whether additional information should be collected.

⇒ Compare the EVPI with the cost of gathering information.

⇒ If the cost of information is greater than the EVPI, then the information should not be collected, because it would waste time and not be cost-effective.

There are three methods to calculate EVPI:

1. Calculation based on profit

$$EVPI = EPPI - EV(\text{optimal})$$

Where  $EPPI$  = the expected payoff when perfect information is available

$$= \sum_{i=1}^n \Pr(E_i) \text{Max}[C_{1i}, C_{2i}, \dots, C_{ni}]$$

$EV(\text{optimal})$  = the expected profit of the best alternative when no additional information is available

**Example 27:** From Example 26, calculate the EVPI.

**Solution**

| Strategy    | Expected Events |       |       | Expected payoff $EV(A_i)$             |
|-------------|-----------------|-------|-------|---------------------------------------|
|             | $E_1$           | $E_2$ | $E_3$ |                                       |
| Strategy A  | 14              | 12    | 1     | $10.2 \Rightarrow EV(\text{optimal})$ |
| Strategy B  | 3               | 3     | 4     | 3.2                                   |
| Strategy C  | 8               | 2     | 9     | 4.6                                   |
| Probability | 0.2             | 0.6   | 0.2   |                                       |

- $\Rightarrow$  Maximum profit 14      12      9  
 $\Rightarrow EPPI = 0.2(14) + 0.6(12) + (0.2)9 = 11.8$  million baht  
 $\Rightarrow EV(\text{optimal}) = 10.2$  million baht  
 $\Rightarrow EVPI = EPPI - EV(\text{optimal}) = 11.8 - 10.2 = 1.6$  million baht  
 $\Rightarrow$  The cost of collecting additional information must not exceed 1.6 million baht.

## 2. Calculation Based on Cost

$EVPI$  = The expected reduction in cost when perfect information is available.

= The expected cost of the best alternative when no information is available.

- The expected cost when perfect information is available.

When the expected cost with perfect information

$$= \sum_{i=1}^n \Pr(E_i) \text{Min}[C_{1i}, C_{2i}, \dots, C_{ni}]$$

The expected cost of the best alternative when no information is available

$$= \text{Min}(EV(A_1), EV(A_2), \dots, EV(A_n))$$

**Example 28:** Vanich Industry Co., Ltd., a manufacturer of electric fans, operates two plants: plant A in Rayong produces fan components, and plant B in Pathum Thani assembles the fans. Normally, Plant A ships components in boxes containing 4,000 pieces each. Based on experience, the manager of Plant B estimates that 5-15% of the parts are defective. The manager at Plant B has **two alternatives**:

1. Inspect every component received from Plant A before assembly. If defective parts are found, an inspection cost of 10 baht per piece is incurred.

2. Do not inspect and assemble directly. If a fan does not function, the defective component will be replaced at a cost of 100 baht per piece.

How should the manager decide, given that the percentage of defective parts may be 5%, 10%, or 15%, with probabilities as shown in the table?

Table showing costs for each alternative (Unit: Ten Thousand Baht)

| Alternative                 | Percentage of defective parts |     |     | Expected Cost without Perfect Information $EV(A_i)$ |
|-----------------------------|-------------------------------|-----|-----|-----------------------------------------------------|
|                             | 5%                            | 10% | 15% |                                                     |
| Inspect every piece         | 4                             | 4   | 4   | $0.5(4) + 0.3(4) + 0.2(4) = 4.0$                    |
| Assemble without inspection | 2                             | 5   | 6   | $0.5(2) + 0.3(5) + 0.2(6) = 3.7$                    |
| Probability                 | 0.5                           | 0.3 | 0.2 |                                                     |

⇒ Without perfect information, the minimum expected cost = 3.7 ten-thousand baht

⇒ Expected cost with perfect information =  $0.5(2) + 0.3(4) + 0.2(4) = 3.0$  ten-thousand baht

⇒  $EVPI = 3.7 - 3.0 = 0.7$  ten-thousand baht or 7,000 baht

\* Therefore, having perfect information would reduce costs by 7,000 baht.

### 3. Calculation Based on Opportunity Loss

$$EVPI = \min(EOL(A_1), EOL(A_2), \dots, EOL(A_n))$$

Where  $EOL(A_i)$  = The expected opportunity loss when choosing alternative  $i$ .

**Example 29** From Example 26, calculate the  $EVPI$  using the opportunity loss table shown below.

### Solution

| Strategy    | Expected Events |       |       | Expected Opportunity Loss $EOL(A_i)$ |
|-------------|-----------------|-------|-------|--------------------------------------|
|             | $E_1$           | $E_2$ | $E_3$ |                                      |
| Strategy A  | 0               | 0     | 8     | $1.6 \Rightarrow \text{Min}$         |
| Strategy B  | 11              | 9     | 5     | 8.6                                  |
| Strategy C  | 6               | 10    | 0     | 7.2                                  |
| Probability | 0.2             | 0.6   | 0.2   |                                      |

$$\Rightarrow EVPI = 1.6 \text{ million baht}$$

It will be found that the  $EVPI$  value from opportunity cost equals the  $EVPI$  value from profit and from cost or expense.

### 6.3.7 Decision tree

#### 6.3.7.1 Constructing a Decision Tree

- Required information
  - Alternative of action.
  - Future events that may occur.
  - The payoff received from each alternative under each event.
  - The probability of occurrence for each event.

- Symbols

○  $\Rightarrow$  Decision node.

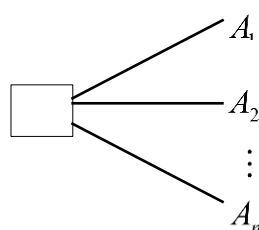
□  $\Rightarrow$  Chance (event) node.

—  $\Rightarrow$  Branches representing decision alternatives or possible events.

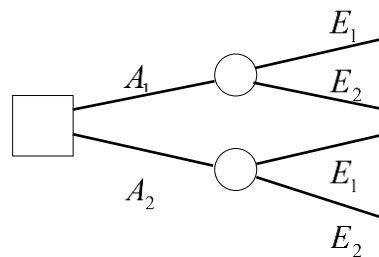
//  $\Rightarrow$  Indicates that a particular branch is not chosen.

- Steps in construction

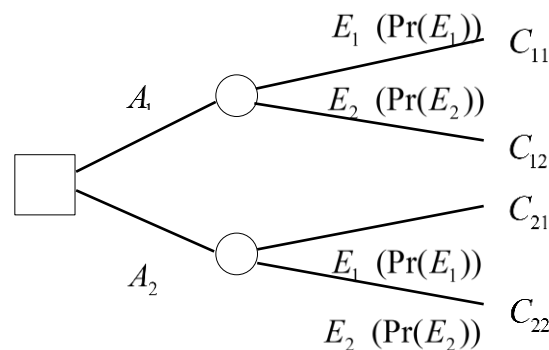
1. Start drawing from left to right, beginning with □ Each branch represents one alternative. If there are  $n$  alternatives ( $A_1, A_2, \dots, A_n$ ).



2. At the end of each alternative branch, add chance nodes representing all possible events that may occur, which is represented by  $\bigcirc$

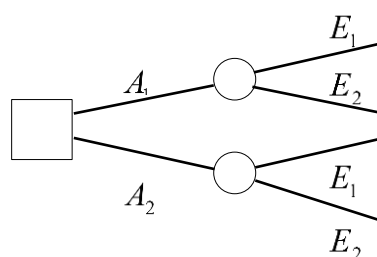


3. At the end of each event branch, specify the payoff value ( $C_{ij}$ ), and label each event branch with its corresponding probability of occurrence.

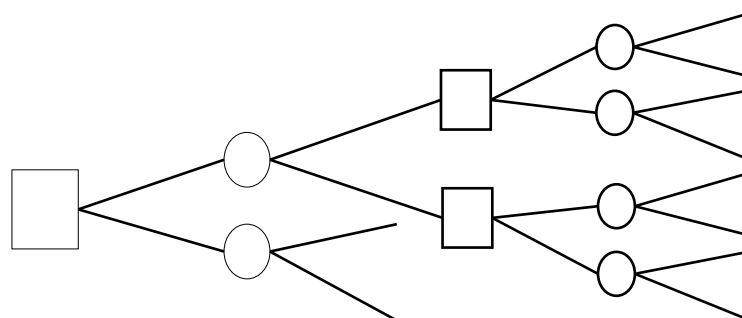


### 6.3.7.2 Types of Decision Trees

1. Decision trees for single-stage decision problems.



2. Decision trees for multi-stage decision problems.



**Example 30:** A ready-to-eat food company has previously sold its products only in the domestic market. The management is considering producing ready-to-eat food for the international market as well, but is uncertain whether the returns would be worthwhile. The company conducted a study and obtained the following information:

**Domestic Market**

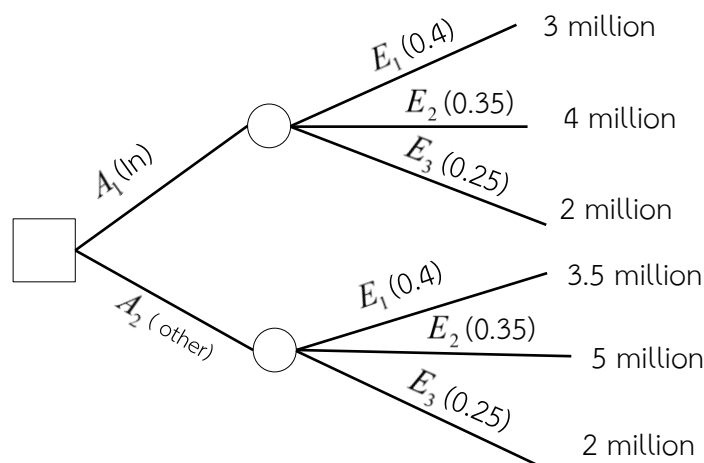
| Market Demand | Probability | Expected Profit |
|---------------|-------------|-----------------|
| Low           | 0.40        | 3 million baht  |
| Medium        | 0.35        | 4 million baht  |
| High          | 0.25        | 2 million baht  |

**International Market**

| Market Demand | Probability | Expected Profit  |
|---------------|-------------|------------------|
| Low           | 0.40        | 3.5 million baht |
| Medium        | 0.35        | 5 million baht   |
| High          | 0.25        | 2 million baht   |

**Solution**

- Decision Maker  $\Rightarrow$  The company's executives.
- Alternatives  $\Rightarrow$  Two options: Domestic market ( $A_1$ ), and International Market ( $A_2$ ).
- Events  $\Rightarrow$  Three possible events: Low demand ( $E_1$ ), Medium demand ( $E_2$ ), and High demand ( $E_3$ ).
- Payoff  $\Rightarrow$  the expected profit to be received.



**Example 31:** New Light Company has found that its product sales have dropped significantly. To address the problem, the company conducted a market study and identified the following alternatives:

1. Conduct research and development to create a new product.
2. Continue producing the existing product.
3. Stop production and discontinue sales

Details of each alternative are as follows:

1. If the company conducts research and development, there is a 75% chance that the research results will be satisfactory. If the results are satisfactory and the company introduces a new product to the market, it is expected that: If sales are high (with a 95% probability), the profit will be 100,000 baht per year. If sales are low, the company will incur a loss of 30,000 baht per year. However, if the company decides not to produce the new product, it will still have to bear the research cost of 15,000 baht.

In the case where the research results are unsatisfactory and the company continues selling the existing product: If sales are high, the profit will be 50,000 baht per year. If sales are low, the loss will be 10,000 baht per year. If the company decides to stop production, it will still incur the research cost of 15,000 baht.

2. If the company does not conduct research and continues selling its existing product, the probability of achieving high sales is only 35%. It is expected that: If sales are high, the profit will be 80,000 baht. If sales are low, the company will incur a loss of 18,000 baht.

3. If the company stops production and discontinues sales, it will incur an inventory holding cost of 5,000 baht per year.

**Solution**

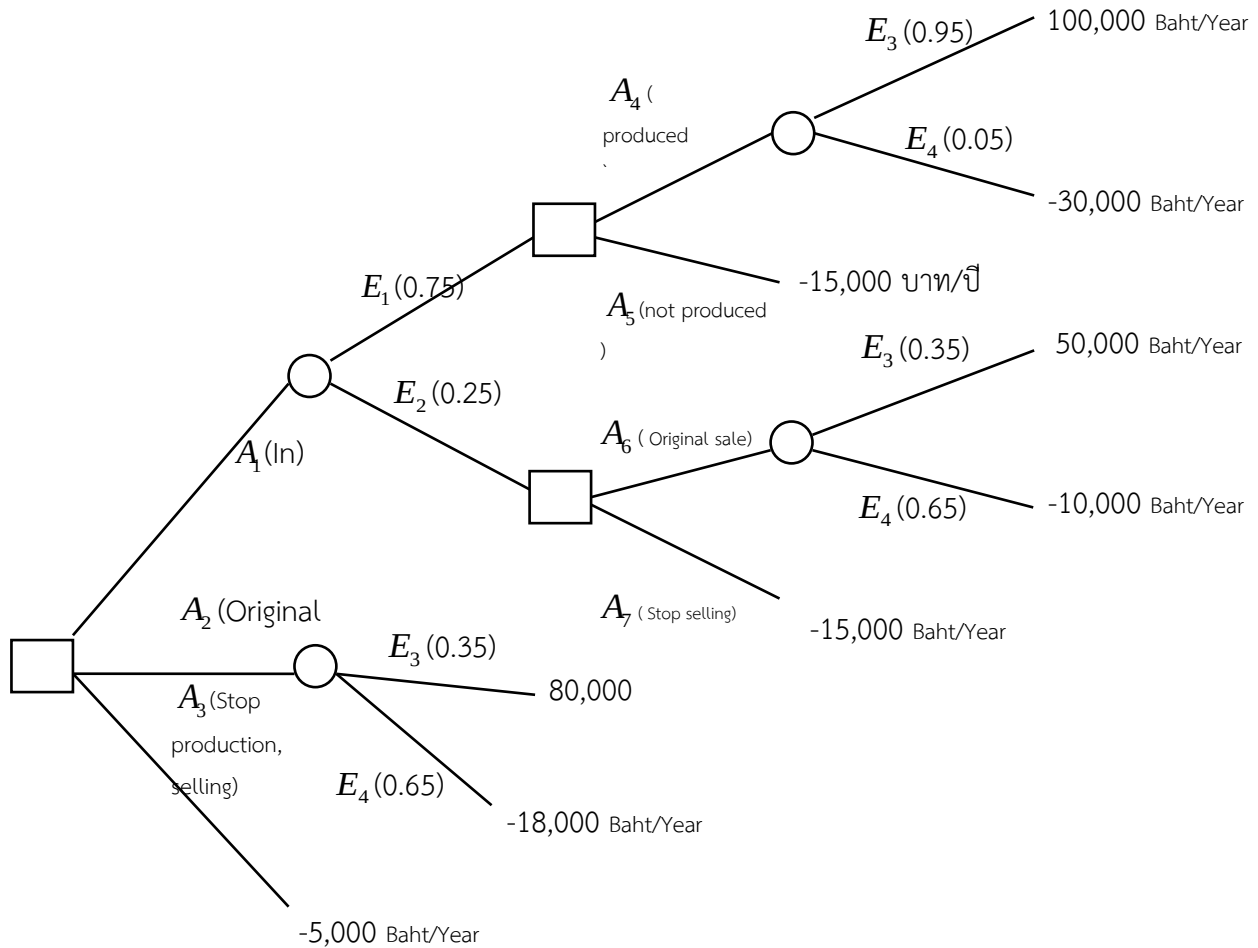
- $A_1$ : Conduct research and development to create a new product
- $A_2$ : Continue producing the existing product
- $A_3$ : Stop production and discontinue sales
- $A_4$ : Producing a new product
- $A_5$ : Do not produce a new product
- $A_6$ : Continue selling the existing product
- $A_7$ : Stop selling the existing product

$E_1$  : Research results are satisfactory

$E_2$  : Research results are unsatisfactory

$E_3$  : High sales

$E_4$  : Low sales

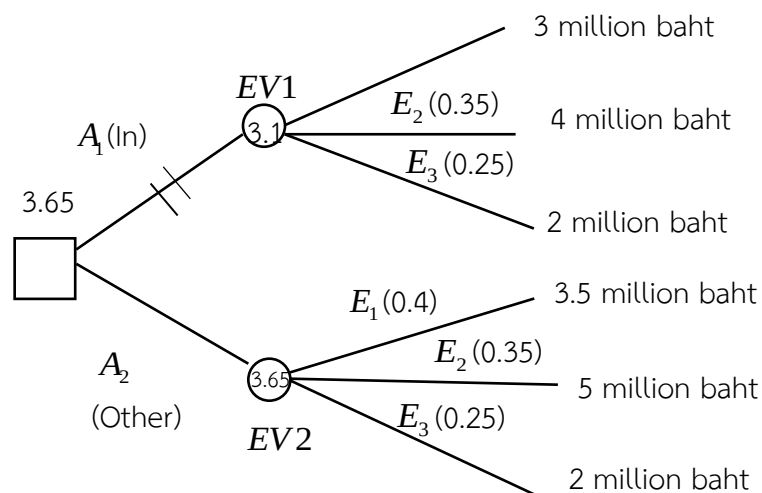


### 6.3.7.3 Analyzing a Decision Tree

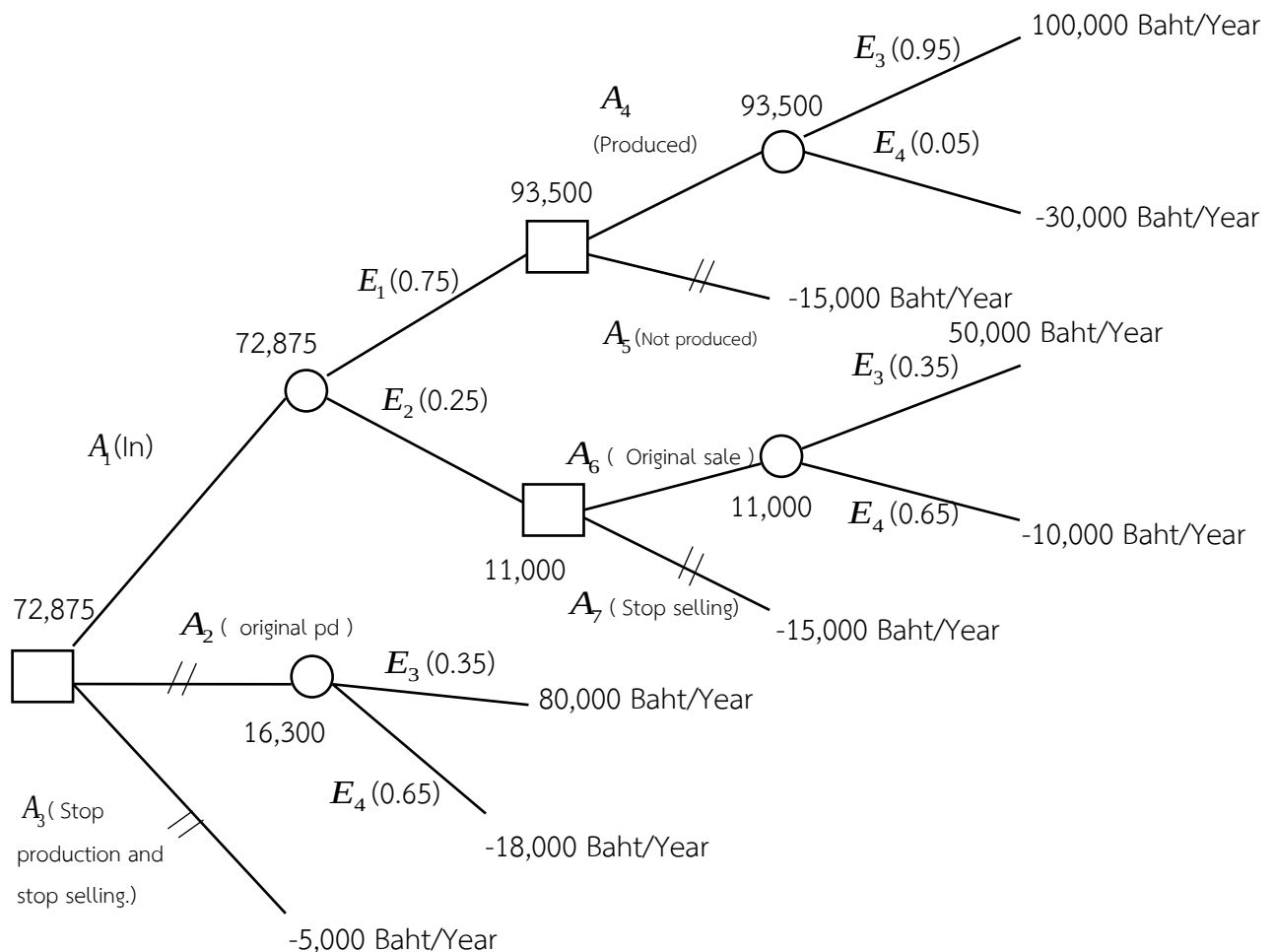
Steps in the analysis:

1. Calculate the expected payoff for each alternative by working from right to left, and write the value at the end of each decision branch.
2. Compare the expected payoffs of all alternatives in order to select the best option.

**Example 32** Based on Example 30, how should the ready-to-eat food company make its decision?



**Example 33:** Based on Example 31, how should New Light Company make its decision?



**Summary** If Alternative A<sub>1</sub>, which is to conduct research and development, is chosen, then if the research results are satisfactory and the company produces the new product, the average expected return will be 93,500 baht, whereas if the research results are unsatisfactory and the company continues selling the existing product, the average expected return will be 11,000 baht.

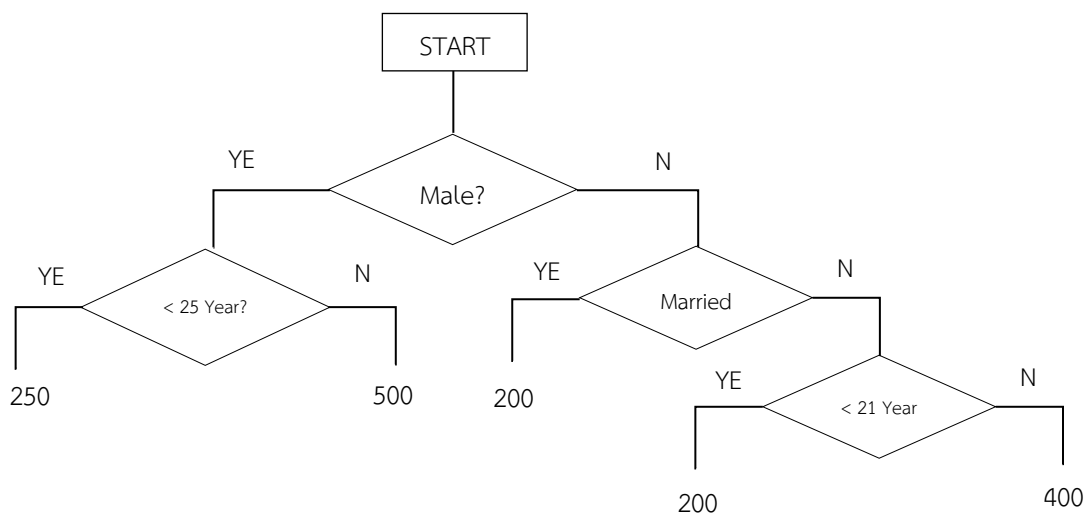
### 6.3.8 Logical Tree

#### Example of an Application

**Example 34** For membership in a certain sunbathing association, men pay a membership fee of 500 baht per year, but if they are under 25 years old, they pay only half that amount. Women pay a membership fee of 400 baht per year, except that those under 21 years old pay only half price; however, any married woman pays only half the membership fee regardless of age.

From this situation, the following questions can be formulated:

1. Are you male?
2. Are you under 25 years old?
3. Are you married?
4. Are you under 21 years old?



#### ● Decision tree and Logical tree

##### Similarities

1. Both use a branching (tree-like) structure.
2. Both are tools for identifying alternatives to solve problems.
3. Both start from a single starting point.
4. Both have multiple possible end points.

##### Differences

Decision tree  $\Rightarrow$  Used to compare which path is best and select one optimal path.

Logical tree  $\Rightarrow$  Not used to choose the best alternative, but to classify or categorize in order to determine the answer for each category.

## 6.4 Decision Making by Updating Information with Sample Data

### 6.4.1 Prior Analysis

This is decision-making that determines the expected payoff by calculating from the prior probabilities currently available.

### 6.4.2 Posterior Analysis

This is decision-making that determines the expected payoff by calculating from posterior probabilities, which are obtained by updating prior probabilities with sample information in order to reduce uncertainty and risk.

#### 6.4.2.1 Bayes Theorem

Let A, B, and C be mutually exclusive events (but collectively exhaustive), and let D be any event that has occurred. At  $\Pr(D) \neq 0$ , the probability of event A given D is:

$$\Pr(A | D) = \frac{\Pr(A) \Pr(D | A)}{\Pr(A) \Pr(D | A) + \Pr(B) \Pr(D | B) + \Pr(C) \Pr(D | C)}$$

#### 6.4.2.2 Posterior probability

Let  $E_1, E_2, E_3$  be any event;  $X$  It is current information (obtained later).

$$\Rightarrow P(E_1), P(E_2), P(E_3)$$

$$\Rightarrow P(X | E_1), P(X | E_2), P(X | E_3)$$

$$\therefore P(E_i | X) = \frac{P(E_i)P(X | E_i)}{P(E_1)P(X | E_1) + P(E_2)P(X | E_2) + P(E_3)P(X | E_3)}$$

**Example 35:** A company producing a brand of instant noodles is deciding among three marketing strategies to increase sales:

Strategy A: Buy 1 Get 1 Free.

Strategy B: A gold prize (1 salung in weight) inside the package.

Strategy C: Collect the letters spelling “MAMA” to receive 1 million baht. The profit to be earned depends on the level of sales, namely  $E_1$  (high),  $E_2$  (medium),  $E_3$  (low) and the payoff table can be constructed as follows:

| Strategy    | Expected Events (Unit:<br>Million Baht) |       |       |
|-------------|-----------------------------------------|-------|-------|
|             | $E_1$                                   | $E_2$ | $E_3$ |
| Strategy A  | 14                                      | 12    | 1     |
| Strategy B  | 3                                       | 3     | 4     |
| Strategy C  | 8                                       | 2     | 9     |
| Probability | 0.2                                     | 0.6   | 0.2   |

Suppose a market research institute surveys consumer satisfaction with the company’s product and obtains three possible outcomes:

$X_1$  : Very satisfied  $X_2$  : Moderately satisfied

$X_3$  : Slightly satisfied

These outcomes correspond to the events  $E_1$ ,  $E_2$ , and  $E_3$ . Based on past data, if the actual sales level is high ( $E_1$ ), there is an 80% chance that consumers will be highly satisfied, which can be written as  $\Pr(X_1 | E_1) = 0.80$ . Similarly, the probability of high consumer satisfaction is 20% given moderate sales ( $\Pr(X_1 | E_2) = 0.20$ ), and 10% given low sales ( $\Pr(X_1 | E_3) = 0.10$ ). These probabilities ( $\sum \Pr(X_i | E_j)$ ) do not have to sum to 1 since they are given under different events. \*However, they must sum to 1 when they are conditional on the same event, meaning that  $\Pr(E_1 | X) + \Pr(E_2 | X) + \Pr(E_3 | X) = 1$

$$\begin{aligned}
 \Rightarrow \Pr(E_1 | X_1) &= \frac{\Pr(E_1) \Pr(X_1 | E_1)}{\Pr(E_1) \Pr(X_1 | E_1) + \Pr(E_2) \Pr(X_1 | E_2) + \Pr(E_3) \Pr(X_1 | E_3)} \\
 &= \frac{(0.2)(0.8)}{(0.2)(0.8) + (0.6)(0.2) + (0.2)(0.1)} \\
 &= 0.53
 \end{aligned}$$

$$\begin{aligned}\Rightarrow \Pr(E_2 | X_1) &= \frac{\Pr(E_2) \Pr(X_1 | E_2)}{\Pr(E_1) \Pr(X_1 | E_1) + \Pr(E_2) \Pr(X_1 | E_2) + \Pr(E_3) \Pr(X_1 | E_3)} \\ &= \frac{(0.6)(0.2)}{(0.2)(0.8) + (0.6)(0.2) + (0.2)(0.1)} \\ &= 0.40 \\ \Rightarrow \Pr(E_3 | X_1) &= \frac{\Pr(E_3) \Pr(X_1 | E_3)}{\Pr(E_1) \Pr(X_1 | E_1) + \Pr(E_2) \Pr(X_1 | E_2) + \Pr(E_3) \Pr(X_1 | E_3)} \\ &= \frac{(0.2)(0.1)}{(0.2)(0.8) + (0.6)(0.2) + (0.2)(0.1)} \\ &= 0.07\end{aligned}$$

It can be found that  $\Rightarrow \Pr(X_1 | E_1) + \Pr(X_1 | E_2) + \Pr(X_1 | E_3) = 0.53 + 0.4 + 0.07 = 1$

### 6.4.3 Decision Making by Updating Information from Sample Data

- Revised expected payoff under uncertainty

$$EV(A_i)_p = \sum_{j=1}^m \Pr(E_j | X_k) C_{ij}$$

Where  $\Pr(E_j | X_k)$  = posterior probability of event  $j$  given sample outcome  $X_k$ .

$C_{ij}$  = profit from alternative  $i$  when event  $j$  occurs.

- Revised expected opportunity loss under uncertainty.

$$EOL(A_i)_p = \sum_{j=1}^m \Pr(E_j | X_k) C_{ij}$$

Where  $C_{ij}$  = opportunity loss from alternative  $i$  when event  $j$  occurs.

**Example 36:** From Example 35, analyze the decision problem by updating the information using sample data.

#### Solution

From the payoff table, the results are as follows (Unit: Million Baht).

| Strategy   | Expected Events |      |      | Expected payoff after updating the information.<br>( $EV(A_i)_p$ ) | $EV(A_i)$ |
|------------|-----------------|------|------|--------------------------------------------------------------------|-----------|
|            | $E1$            | $E2$ | $E3$ |                                                                    |           |
| Strategy A | 14              | 12   | 1    | 12.29                                                              | 10.2      |

|                  |      |      |      |      |     |
|------------------|------|------|------|------|-----|
| Strategy B       | 3    | 3    | 4    | 3.07 | 3.4 |
| Strategy C       | 8    | 2    | 9    | 5.67 | 4.6 |
| $\Pr(E_j   X_1)$ | 0.53 | 0.40 | 0.07 |      |     |

$$\Rightarrow EV(A_1) = 14(0.53) + 12(0.4) + 1(0.07) = 12.29$$

$\Rightarrow$  Choose Strategy A

**Example 37** A construction company is considering bidding on a building project. The outcome of the project depends on the construction workers. If the workers go on strike, the company would earn a profit of about 500,000 baht from the project. On the other hand, if the workers do not go on strike, the profit would be 1,000,000 baht. The company manager believes that the probability of a strike is 20%. He is unsure whether to accept this project, since he already has another construction job that would provide a certain profit of 875,000 baht.

After consulting experienced individuals, he concluded that inflation and labor strikes are closely related. It is estimated that if a strike occurs, the probability of inflation is 0.7, and if there is no strike, the probability of inflation is 0.375. How should the company manager make his decision when:

1. Relying on the existing information?
2. Updating the existing information using data from experienced sources?

**Solution**

- Decision Maker  $\Rightarrow$  The company manager
- Alternatives  $\Rightarrow$  Two options: accept the construction project ( $A_1$ ), do not accept the project ( $A_2$ ).
- Events  $\Rightarrow$  Two possible events: workers go on strike ( $E_1$ ), workers do not go on strike ( $E_2$ ).
- Payoff  $\Rightarrow$  Profit earned.

1. Using the existing information (prior analysis)

The payoff table can be constructed as follows (unit: thousand baht):

| Events                      | $P(E_j)$ | $P(E_j   X)$ | Alternatives             |                                 |
|-----------------------------|----------|--------------|--------------------------|---------------------------------|
|                             |          |              | Accept Project ( $A_1$ ) | Do Not Accept Project ( $A_2$ ) |
| Workers do not strike $E_1$ | 0.8      | 0.68         | 1,000                    | 875                             |
| Workers strike $E_2$        | 0.2      | 0.32         | 500                      | 875                             |

$$\Rightarrow EPPI_0(\text{optimal}) = 1000(0.8) + 875(0.2) = 975 \Rightarrow \text{thousand baht}$$

$$EV_0(\text{optimal}) = 900 \text{ thousand baht}$$

$$\Rightarrow EVPI_0(\text{optimal}) = EPPI_0 - EV_0(\text{optimal}) = 975 - 900 = 75 \text{ thousand baht or } 75,000 \text{ baht.}$$

Choose  $\Rightarrow A_1$

## 2. Updating the existing information using data from experienced sources (Posterior Analysis)

Let inflation be the event.  $\Rightarrow \Pr(X | E_1) = 0.375, \Pr(X | E_2) = 0.70$

Using this sample information, the prior probabilities can be updated as follows:

$$\begin{aligned} \Rightarrow \Pr(E_1 | X) &= \frac{\Pr(E_1) \Pr(X | E_1)}{\Pr(E_1) \Pr(X | E_1) + \Pr(E_2) \Pr(X | E_2)} \\ &= \frac{(0.8)(0.375)}{(0.8)(0.375) + (0.2)(0.7)} \\ &= 0.68 \end{aligned}$$

$$\begin{aligned} \Rightarrow \Pr(E_2 | X) &= \frac{\Pr(E_2) \Pr(X | E_2)}{\Pr(E_1) \Pr(X | E_1) + \Pr(E_2) \Pr(X | E_2)} \\ &= \frac{(0.2)(0.7)}{(0.8)(0.375) + (0.2)(0.7)} \\ &= 0.32 \end{aligned}$$

$$\Rightarrow EV(A_1)p = (0.68)(1000) + (0.32)(500) = 840$$

$$\Rightarrow EV(A_2)p = (0.68)(875) + (0.32)(875) = 875 \Rightarrow \text{Max}$$

Choose  $\Rightarrow A_2$ : not to accept the construction project.

The value of the information is:  $EVPIp = EPPIp - EVp(\text{optimal})$

$$\Rightarrow EPPIp = (0.68)(1,000) + (0.32)(875) = 960 \text{ thousand baht.}$$

$$EVp(\text{optimal}) = 875 \text{ thousand baht.}$$

$$\Rightarrow EVPIp = 960 - 875 = 85 \text{ thousand baht.}$$

It can be observed that the value of information increases, indicating that the information used (“inflation”) from the beginning may not be appropriate (and may not actually be related to labor strikes). In reality, the value of information should decrease when it is less relevant.

#### 6.4.4 Decision Making When Sample Information Follows a Binomial Distribution

**Example 38** A factory wants to check the quality of its products before delivering them to customers. Historically, the defect rates are 3%, 5%, and 10% with probabilities 0.50, 0.30, and 0.20, respectively. The factory will ship the products if the defect rate is 3%. The loss analysis is shown in the table below.

| % Defective | Pr( $E_j$ ) | Alternatives (Unit: million baht) |                   |
|-------------|-------------|-----------------------------------|-------------------|
|             |             | Do not deliver ( $A_1$ )          | Deliver ( $A_2$ ) |
| 3%          | 0.50        | 200                               | 0                 |
| 5%          | 0.30        | 0                                 | 100               |
| 10%         | 0.20        | 0                                 | 200               |

- Analyze the problem using prior information (prior analysis).
- If a random sample of 10 items is inspected and 2 defectives are found, analyze the problem using this sample information together with the prior information.

#### Solution

- Alternatives  $\Rightarrow$  Two options: Do not deliver ( $A_1$ ), Deliver ( $A_2$ ).
- Events  $\Rightarrow$  Three possible defect rates: 3% ( $E_1$ ), 5% ( $E_2$ ), 10% ( $E_3$ ).
- Payoff  $\Rightarrow$  Loss value.

- Using existing information (prior analysis)

Construct the loss table as follows (unit: million baht):

| Events | Alternatives (unit: million baht) |
|--------|-----------------------------------|
|--------|-----------------------------------|

|       | $\Pr(E_j)$ | Do not deliver ( $A_1$ ) | Deliver ( $A_2$ ) |
|-------|------------|--------------------------|-------------------|
| $E_1$ | 0.50       | 200                      | 0                 |
| $E_2$ | 0.30       | 0                        | 100               |
| $E_3$ | 0.20       | 0                        | 200               |
| $A_2$ |            | 100                      | 70                |

$\Rightarrow$  Choose  $A_2$  : Deliver the products and got  $EVPI = 70 = EOL$

(lowest).

b. Updating information using sample inspection (Posterior Analysis)

Let  $x$  = number of defective items found in a random sample of 10 products.

$\Rightarrow X \sim \text{Binomial}(n=10, p)$ ;  $p$  = % defective items =  $E_j$ .

$$\Rightarrow P(X = 2 | p = E_j) = \binom{10}{2} E_j^2 (1 - E_j)^{10-2}$$

Using the binomial distribution, update the prior probabilities:

| Events ( $\Pr(E_j   X = 2)$ ) | $\Pr(E_j)$ | $\Pr(X = 2   E_j)$ | $\Pr(E_j) \times \Pr(X = 2   E_j)$ | $\Pr(E_j   X = 2)$      |
|-------------------------------|------------|--------------------|------------------------------------|-------------------------|
| 0.03                          | 0.50       | 0.032              | 0.0160                             | $0.0160/0.0773 = 0.207$ |
| 0.05                          | 0.30       | 0.075              | 0.0225                             | $0.0225/0.0773 = 0.291$ |
| 0.10                          | 0.20       | 0.194              | 0.0388                             | $0.0388/0.0773 = 0.502$ |
| Total                         |            |                    | 0.0773                             | 1.000                   |

$$\Pr(X = 2 | E_1) = \binom{10}{2} (0.03)^2 (1 - 0.03)^{10-2} = 0.032$$

$$\Rightarrow EOL(A_1)p = (0.21)(200) + (0.29)(0) + 0.50(0) = 42.0 \text{ Min}$$

$$\Rightarrow EOL(A_2)p = (0.21)(0) + (0.29)(100) + 0.50(200) = 129.0$$

$\Rightarrow$  Choose  $A_1$  : Do not deliver the products and got  $EVPI = 42$

In this case, the value in part a) ( $EOL = 70$ ) and the value in part b) ( $EOL = 42$ ) indicate that alternative (b) should be chosen, that is, not delivering the products is better because it results in a lower expected loss.

But if the value in part a) ( $EOL = 70$ ) and the value in part b) ( $EOL = 68$ ) are very close, it means the results are nearly the same. Therefore, sampling inspection for defects would not be necessary, and the products could be delivered right away.

### 6.4.5 Decision Making When Sample Information Follows a Poisson Distribution

**Example 39** From Example 38, suppose a random sample of 50 items is inspected and 5 defective products are found. Analyze the problem using this sample information together with the prior information.

#### Solution

Let  $X$  = the number of defectives found in a sample of 50 items be given,

$\Rightarrow X \sim \text{Binomial}(n=50, p)$ ;  $p$  = the defect proportion = 0.03, 0.05, 0.10

Since  $n > 25$  and  $np < 5$

The Binomial distribution can be approximated by the Poisson distribution.

$$P(X = 5 | \lambda_j = 50E_j) = \frac{e^{-\lambda_j} \lambda_j^5}{5!}$$

Using this sample information to update the prior probabilities:

| $E_j$ | $\Pr(E_j)$ | $\lambda_j = 50 \times E_j$ | $\Pr(X = 5   \lambda_j)$ | $\Pr(E_j) \times \Pr(X = 5   \lambda_j)$ | $\Pr(\lambda_j   X = 2)$ |
|-------|------------|-----------------------------|--------------------------|------------------------------------------|--------------------------|
| 0.03  | 0.50       | 1.5                         | 0.01                     | 0.005                                    | 0.005/0.062<br>=0.08     |
| 0.05  | 0.30       | 2.5                         | 0.07                     | 0.021                                    | 0.021/0.062<br>=0.34     |
| 0.10  | 0.20       | 5.0                         | 0.18                     | 0.036                                    | 0.036/0.062<br>=0.58     |
| Total |            |                             |                          | 0.062                                    | 1.00                     |

$$\Pr(X = 5 | \lambda_1) = \frac{e^{-1.5} 1.5^5}{5!} = 0.01$$

$$\Rightarrow EOL(A_1)p = (0.08)(200) + (0.34)(0) + 0.58(0) = 16.0 \quad \text{Min}$$

$$EOL(A_2)p = (0.08)(0) + (0.34)(100) + 0.58(200) = 150.0$$

$\Rightarrow$  Choose  $A_1$  : Do **not** deliver the products and get  $EVPI = 16$

Therefore, it is advisable to conduct sampling inspection before deciding whether to deliver or not, as this reduces expected losses.  $\Rightarrow$  In general, increasing the sample size  $n$  leads to lower expected loss and reduced decision risk.